CLASS 10 SCIENCE: CHEMICAL REACTIONS & EQUATIONS

Complete Worksheet Questions with Detailed Solutions & Explanations

SECTION A: MULTIPLE CHOICE QUESTIONS

Question 1: $MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2$
Identify the substance oxidized in the above equation.

(a) $MnCl_2$ (b) HCl (c) H_2O (d) MnO_2

Answer: (b) HCl

Explanation: Oxidation involves the loss of hydrogen or addition of oxygen. In this reaction, HCl loses hydrogen to form Cl_2, so HCl is oxidised.

Question 2: Electrolysis of water is a decomposition reaction. The mole ratio of hydrogen and oxygen gases liberated during electrolysis of water is:

(a) 1 : 1 (b) 2 : 1 (c) 4 : 1 (d) 1 : 2

Answer: (b) 2 : 1

Explanation: The balanced equation for electrolysis of water is $2H_2O(l) \rightarrow 2H_2(g) + O_2(g)$. Two moles of hydrogen gas are produced for every 1 mole of oxygen gas, giving a mole ratio of 2:1.

Question 3: $Pb + CuCl_2 \rightarrow PbCl_2 + Cu$
The above reaction is an example of:

(a) combination (b) double displacement (c) decomposition (d) displacement

Answer: (d) displacement

Explanation: Lead (Pb) is more reactive than copper (Cu). It displaces copper from copper chloride solution, forming lead chloride and copper metal. This is a single displacement reaction.

Question 4: When green coloured ferrous sulphate crystals are heated, the colour of the crystal changes because:

(a) it is decomposed to ferric oxide (b) it loses water of crystallisation (c) it forms SO_2 (d) it forms SO_3

Answer: (b) it loses water of crystallisation

Explanation: Hydrated ferrous sulphate crystals ($FeSO_4 \cdot 7H_2O$) are green. Upon heating, they lose their water of crystallisation to form anhydrous ferrous sulphate ($FeSO_4$), which causes the colour to fade/change.

Question 5: $2HNO_3 + Ca(OH)_2 \rightarrow Ca(NO_3)_2 + 2H_2O$ is an example of:
(i) displacement reaction (ii) double displacement reaction
(iii) neutralisation reaction (iv) combination reaction.

(a) (i) and (ii) (b) (ii) and (iii) (c) (iii) and (iv) (d) (i) and (iv)

Answer: (b) (ii) and (iii)

Explanation: It is a double displacement reaction because there is a mutual exchange of ions between reactants. It is also a neutralisation reaction because an acid (HNO_3) reacts with a base [$Ca(OH)_2$] to form salt and water.

Question 6: What is observed when a solution of potassium iodide is added to silver nitrate solution?

(a) No reaction takes place (b) White precipitate of silver iodide is formed (c) yellow precipitate of AgI is formed (d) AgI is soluble in water

Answer: (c) yellow precipitate of AgI is formed

Explanation: When potassium iodide (KI) reacts with silver nitrate ($AgNO_3$), a double displacement reaction occurs: $KI(aq) + AgNO_3(aq) \rightarrow AgI(s)\downarrow + KNO_3(aq)$. Silver iodide (AgI) is insoluble and forms a yellow precipitate.

Question 7: Identify 'x', 'y' and 'z' in the following balanced reaction:

$x\ Pb(NO_3)_2(s) \rightarrow y\ PbO(s) + z\ NO_2(g) + O_2(g)$

(a) 2, 4, 2 (b) 2, 2, 4 (c) 2, 4, 4 (d) 4, 2, 2

Answer: (b) 2, 2, 4

Explanation: The balanced equation for the thermal decomposition of lead nitrate is: $2Pb(NO_3)_2(s) \rightarrow 2PbO(s) + 4NO_2(g) + O_2(g)$. Comparing coefficients gives x = 2, y = 2, and z = 4.

Question 8: Oxidation involves:

(i) gain of electron (ii) loss of electron

(iii) addition of oxygen or electronegative element (iv) removal of hydrogen or electropositive element

(a) (i), (ii), (iii) (b) (ii), (iii), (iv) (c) (i), (iii), (iv) (d) (i), (ii), (iv)

Answer: (b) (ii), (iii), (iv)

Explanation: Oxidation is defined as the loss of electrons, addition of oxygen or an electronegative element, or removal of hydrogen or an electropositive element. Gain of electrons represents reduction.

Question 9: Which of the following statements about the given reaction are correct?

$3Fe(s) + 4H_2O(g) \rightarrow Fe_3O_4(s) + 4H_2(g)$

(i) Iron metal is getting oxidised (ii) Water is getting reduced

(iii) Water is acting as reducing agent (iv) Water is acting as an oxidising agent

(a) (i), (ii) and (iii) (b) (iii) and (iv) (c) (i), (ii) and (iv) (d) (ii) and (iv)

Answer: (c) (i), (ii) and (iv)

Explanation: Iron (Fe) gains oxygen to form Fe_3O_4, so it is oxidised. Water (H_2O) loses oxygen to form H_2, so it is reduced. Because water provides oxygen to oxidise iron, water acts as the oxidising agent.

Question 10: Barium chloride on reacting with ammonium sulphate forms barium sulphate and ammonium chloride. Which of the following correctly represents the type of the reaction involved?

(i) Displacement reaction (ii) Precipitation reaction

(iii) Combination reaction (iv) Double displacement reaction

(a) (i) only (b) (ii) only (c) (iv) only (d) (ii) and (iv)

Answer: (d) (ii) and (iv)

Explanation: The reaction $BaCl_2(aq) + (NH_4)_2SO_4(aq) \rightarrow BaSO_4(s)\downarrow + 2NH_4Cl(aq)$ involves exchange of ions between two compounds (double displacement) and forms an insoluble white precipitate of $BaSO_4$ (precipitation).

SECTION B: ASSERTION-REASON TYPE QUESTIONS

DIRECTION: Each of these questions contains an Assertion (A) followed by Reason (R). Read them carefully and select the correct option from the choices below:
(a) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
(b) Both Assertion and Reason are correct, but Reason is NOT the correct explanation of Assertion.
(c) Assertion is correct but Reason is incorrect.
(d) Assertion is incorrect but Reason is correct.

Question 1: Assertion (A): Photosynthesis is considered as an endothermic reaction.
Reason (R): Energy gets released in the process of photosynthesis.

Answer: (c) Assertion is correct but Reason is incorrect.

Explanation: Photosynthesis absorbs sunlight (energy) to convert carbon dioxide and water into glucose, making it an endothermic process. Thus, Assertion is correct. However, energy is absorbed during photosynthesis, not released, so Reason is incorrect.

Question 2: Assertion (A): Decomposition of vegetable matter into compost is an example of exothermic reactions.
Reason (R): Exothermic reactions are those reactions in which heat is evolved.

Answer: (a) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.

Explanation: Decomposition of organic matter by microbes releases energy in the form of heat, making it an exothermic reaction as defined in the Reason.

Question 3: Assertion (A): When HCl is added to zinc granules, a chemical reaction occurs.
Reason (R): Evolution of a gas indicates that the chemical reaction is taking place.

Answer: (a) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.

Explanation: When dilute HCl reacts with zinc granules, hydrogen gas is liberated ($Zn + 2HCl \rightarrow ZnCl_2 + H_2\uparrow$). Evolution of gas is a characteristic observation confirming that a chemical reaction has occurred.

Question 4: Assertion (A): Calcium carbonate when heated gives calcium oxide and carbon dioxide.
Reason (R): On heating calcium carbonate, decomposition reaction takes place.

Answer: (a) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.

Explanation: Heating calcium carbonate causes thermal decomposition into calcium oxide and carbon dioxide gas ($CaCO_3 \rightarrow CaO + CO_2$). Reason correctly explains why this chemical change takes place.

Question 5: Assertion (A): Chemical reaction changes the physical and chemical properties of a substance.
Reason (R): Chemical change involves a change in the chemical composition of matter, and a new substance is formed.

Answer: (a) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.

Explanation: During a chemical reaction, chemical bonds break and new bonds form to produce new substances with entirely different chemical compositions and physical/chemical properties.

Question 6: Assertion (A): AgBr is used in photographic and X-ray film.
Reason (R): AgBr is photosensitive and changes to Ag and bromine in presence of sunlight and undergoes decomposition reaction.

Answer: (a) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.

Explanation: Silver bromide undergoes photolytic decomposition in the presence of sunlight ($2AgBr \rightarrow 2Ag + Br_2$). This light-sensitive property makes it suitable for photographic and X-ray film emulsions.

Question 7: Assertion (A): Magnesium ribbon keeps on burning in atmosphere of nitrogen.
Reason (R): Magnesium reacts with nitrogen to form magnesium nitrides and this reaction is combination reaction.

Answer: (a) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.

Explanation: Magnesium reacts directly with nitrogen gas to form magnesium nitride ($3Mg + N_2 \rightarrow Mg_3N_2$). Since two elements combine to form a single compound, it is a combination reaction that sustains burning.

Question 8: Assertion (A): Zinc reacts with sulphuric acid to form zinc sulphate and hydrogen gas and it is displacement reaction.
Reason (R): Zinc reacts with oxygen to form zinc oxide.

Answer: (b) Both Assertion and Reason are correct, but Reason is NOT the correct explanation of Assertion.

Explanation: Both statements are individually correct facts. However, the reaction of zinc with oxygen has no connection to why zinc displacing hydrogen from sulphuric acid is a displacement reaction.

Question 9: Assertion (A): $MnO_2 + 4HCl \rightarrow MnCl_2 + Cl_2 + 2H_2O$ is redox reaction.
Reason (R): MnO_2 oxidises HCl to Cl_2 and gets reduced to $MnCl_2$.

Answer: (a) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.

Explanation: In this reaction, MnO_2 causes the oxidation of HCl to Cl_2 while itself getting reduced to $MnCl_2$. The simultaneous oxidation and reduction process defines it as a redox reaction.

Question 10: Assertion (A): White silver chloride turns grey in sunlight.
Reason (R): Decomposition of silver chloride in presence of sunlight takes place to form silver metal and chlorine gas.

Answer: (a) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.

Explanation: White silver chloride undergoes photolytic decomposition in sunlight to yield grey silver metal and chlorine gas ($2AgCl \rightarrow 2Ag + Cl_2$).

SECTION C: CASE STUDY QUESTIONS

Read the following passage and answer the questions that follow:

Reactions in which one element takes place of another element in a compound, are known as displacement reactions. In general, a more reactive element displaces a less reactive element from its compound. In all single displacement reactions, only one element displaces another element from its compound. The displacement reaction between iron (III) oxide and powdered aluminium produces so much heat that iron metal obtained is in molten form.

Question (i): Copper displaces which of the following metals from its salt solution?

(a) $ZnSO_4$ (b) $FeSO_4$ (c) $AgNO_3$ (d) $NiSO_4$

Answer: (c) $AgNO_3$

Explanation: Copper lies above silver (Ag) in the reactivity series, but below zinc, iron, and nickel. Therefore, copper can only displace silver from its nitrate solution: $Cu + 2AgNO_3 \rightarrow Cu(NO_3)_2 + 2Ag$.

Question (ii): When zinc reacts with dilute sulphuric acid, the gas evolved is:

(a) red in colour and have a sweet smelling
(b) green in colour and have a foul smell
(c) colourless, odourless and burns with a pop sound
(d) colourless, pungent smelling and burns with a pop sound

Answer: (c) colourless, odourless and burns with a pop sound

Explanation: Zinc reacts with dilute sulphuric acid to yield hydrogen gas: $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2\uparrow$. Hydrogen gas is colourless, odourless, and burns with a characteristic pop sound.

Question (iii): When dry hydrogen is passed over a heated oxide of metal X using standard apparatus, a reddish-brown residue is obtained. The reddish-brown residue could be:

(a) copper (b) lead (c) silver (d) zinc

Answer: (a) copper

Explanation: Hydrogen gas reduces heated black copper(II) oxide (CuO) to reddish-brown copper metal: $CuO(s) + H_2(g) \rightarrow Cu(s) + H_2O(g)$.

Question (iv): Which of the following reactions is a displacement reaction?

(a) $CaO + H_2O \rightarrow Ca(OH)_2$ (b) $MgCO_3 \rightarrow MgO + CO_2$
(c) $Mg + CuSO_4 \rightarrow MgSO_4 + Cu$ (d) $H_2 + Cl_2 \rightarrow 2HCl$

Answer: (c) $Mg + CuSO_4 \rightarrow MgSO_4 + Cu$

Explanation: In $Mg + CuSO_4 \rightarrow MgSO_4 + Cu$, magnesium displaces copper from copper sulphate solution. Option (a) and (d) are combination reactions, and (b) is a decomposition reaction.

Question (v): When dilute hydrochloric acid is added to granulated zinc placed in a test tube, the observation made is:

(a) the surface of the metal turns shining (b) the reaction mixture turns milky
(c) greenish yellow gas is evolved (d) the colourless and odourless gas evolves with a pop sound.

Answer: (d) the colourless and odourless gas evolves with a pop sound.

Explanation: Granulated zinc reacts with dilute HCl to produce hydrogen gas ($Zn + 2HCl \rightarrow ZnCl_2 + H_2\uparrow$), which is colourless, odourless, and ignites with a pop sound.

SECTION D: SHORT ANSWER TYPE QUESTIONS

Question 1: List four observations that help us to determine whether a chemical reaction has taken place. [Board Term-I, 2012]

Answer: (i) Evolution of a gas
(ii) Change in temperature
(iii) Change in state
(iv) Change in colour

Explanation: These four key physical and chemical changes indicate that old bonds are breaking and new substances with different chemical properties are forming.

Question 2: What is observed when carbon dioxide gas is passed through lime water:
(i) For a short duration
(ii) For long duration?
Also, write the chemical equations for the reactions involved. [Board Term I, 2016]

Answer: (i) For short duration: Limewater turns milky due to the formation of an insoluble precipitate of calcium carbonate ($CaCO_3$).
Equation: $Ca(OH)_2(aq) + CO_2(g) \rightarrow CaCO_3(s)\downarrow + H_2O(l)$

(ii) For long duration: Milkiness disappears and a clear solution is obtained due to the formation of water-soluble calcium hydrogen carbonate [$Ca(HCO_3)_2$].
Equation: $CaCO_3(s) + H_2O(l) + CO_2(g) \rightarrow Ca(HCO_3)_2(aq)$

Explanation: Excess carbon dioxide gas reacts with insoluble calcium carbonate to form soluble calcium bicarbonate, clearing the milky appearance.

Question 3: Write the balanced chemical equation for the following reaction and write the name of the reaction:
Barium Chloride + Aluminium Sulphate → Barium Sulphate + Aluminium Chloride

Answer: Balanced Chemical Equation: $3BaCl_2(aq) + Al_2(SO_4)_3(aq) \rightarrow 3BaSO_4(s)\downarrow + 2AlCl_3(aq)$
Name of Reaction: Double Displacement Reaction (also a Precipitation Reaction).

Explanation: In this reaction, barium ions (Ba^{2+}) and aluminium ions (Al^{3+}) exchange their anions to form an insoluble white precipitate of barium sulphate ($BaSO_4$).

Question 4: Why do we store silver chloride in dark coloured bottles? Explain in brief. [CBSE 2010]

Answer: Silver chloride (AgCl) decomposes into silver metal and chlorine gas when exposed to light: $2AgCl(s) \rightarrow 2Ag(s) + Cl_2(g)$. Therefore, it is stored in dark-coloured bottles.

Explanation: Dark-coloured bottles absorb light rays and prevent light from entering, thereby protecting silver chloride from photolytic decomposition.

Question 5: Define a combination reaction. Give one example of a combination reaction which is also exothermic.

Answer: Definition: A reaction in which two or more reactants combine to form a single product is called a combination reaction.

Exothermic Example: Slaking of lime (reaction of quicklime with water):
$CaO(s) + H_2O(l) \rightarrow Ca(OH)_2(aq) + Heat$

Explanation: Quicklime (CaO) and water combine to form a single product, calcium hydroxide [$Ca(OH)_2$], while liberating a large amount of heat energy.

Question 6: What happens when an iron nail is put inside a copper sulphate solution? Write a reaction with observation.

Answer: Observation: The blue colour of the copper sulphate solution fades and changes to light green. A reddish-brown deposit of copper metal forms on the surface of the iron nail.

Chemical Equation: $Fe(s) + CuSO_4(aq)$ [Blue] $\rightarrow FeSO_4(aq)$ [Light Green] $+ Cu(s)$ [Reddish-brown]

Explanation: Iron is more reactive than copper and displaces copper from copper sulphate solution to form ferrous sulphate and copper metal.

Question 7: Name the raw materials that are required for the manufacture of washing soda by Solvay process. Describe the chemical reactions involved in the process. [2007]

Answer: Raw Materials Required: Sodium chloride (NaCl), Water (H_2O), Ammonia (NH_3), and Limestone ($CaCO_3$, source of CO_2).

Chemical Reactions Involved:
1. Formation of Sodium Bicarbonate:
$NaCl + H_2O + CO_2 + NH_3 \rightarrow NH_4Cl + NaHCO_3$

2. Thermal Decomposition of Sodium Bicarbonate:
$2NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2$

3. Recrystallisation of Sodium Carbonate:
$Na_2CO_3 + 10H_2O \rightarrow Na_2CO_3 \cdot 10H_2O$ (Washing Soda)

Explanation: The Solvay process converts common salt into sodium carbonate decahydrate (washing soda) through intermediate formation and heating of sodium bicarbonate.

Question 8: (i) State the law which is followed in balancing a chemical equation.
(ii) Balance the following chemical equation: $Fe + H_2O \rightarrow Fe_3O_4 + H_2$ [Board Term-I, 2012]

Answer: (i) Law Followed: Law of Conservation of Mass, which states that mass can neither be created nor destroyed in a chemical reaction. Hence, total mass of reactants must equal total mass of products.

(ii) Balanced Chemical Equation: $3Fe(s) + 4H_2O(g) \rightarrow Fe_3O_4(s) + 4H_2(g)$

Explanation: Balancing ensures that the number of atoms of each element (3 Iron, 8 Hydrogen, 4 Oxygen) remains equal on both sides of the equation.

Question 9: A white solid when dropped in water produces a hissing sound. What may the solid be? Give the chemical reaction for the above. Name the product formed. [2010]

Answer: White Solid: Quicklime / Calcium oxide (CaO).

Chemical Reaction: $CaO(s) + H_2O(l) \rightarrow Ca(OH)_2(aq) + Heat$

Product Formed: Calcium hydroxide / Slaked lime [$Ca(OH)_2$].

Explanation: The reaction of quicklime with water is highly exothermic. The rapid evolution of heat produces steam bubbles that create a characteristic hissing sound.

Question 10: When hydrogen gas is passed over heated copper (II) oxide, copper and steam are formed. Write the balanced chemical equation with physical states for this reaction. State what kind of chemical reaction this is. [Board Term-I, 2015]

Answer: Balanced Chemical Equation: $CuO(s) + H_2(g) \rightarrow Cu(s) + H_2O(g)$

Type of Reaction: Redox Reaction (and Displacement Reaction).

Explanation: Copper(II) oxide loses oxygen to form copper metal (reduction), while hydrogen gas gains oxygen to form water vapour/steam (oxidation).

SECTION E: LONG ANSWER QUESTIONS

Question 1: (a) 2g of ferrous sulphate crystals are heated in a dry boiling tube.

(i) List any two observations.

(ii) Name the type of chemical reaction taking place.

(iii) Write the chemical equation for the reaction.

(b) Write the chemical equation of the reaction in which the following changes have taken place with an example of each:

(i) Change in colour (ii) Change in temperature (iii) Formation of precipitate

Answer: (a)

(i) Observations:

- **Green colour of $FeSO_4 \cdot 7H_2O$ crystals fades and a reddish-brown solid (Fe_2O_3) is formed.**
- **Pungent gas with the characteristic smell of burning sulphur (SO_2 and SO_3) is evolved.**

(ii) Type of reaction: Thermal Decomposition Reaction.

(iii) Chemical Equation: $2FeSO_4(s) \rightarrow Fe_2O_3(s) + SO_2(g) + SO_3(g)$

(b)

(i) Change in colour:

$Cu(s) + 2AgNO_3(aq) \rightarrow Cu(NO_3)_2(aq) + 2Ag(s)$

(Colourless $AgNO_3$ solution turns blue due to formation of copper nitrate).

(ii) Change in temperature:

$NaOH(aq) + HCl(aq) \rightarrow NaCl(aq) + H_2O(l) + Heat$

(Neutralisation reaction releases heat, causing temperature to rise).

(iii) Formation of precipitate:

$Pb(NO_3)_2(aq) + 2KI(aq) \rightarrow PbI_2(s)\downarrow + 2KNO_3(aq)$

(Bright yellow precipitate of lead iodide, PbI_2, is formed).

Explanation: This comprehensive question tests key observable criteria of chemical reactions (decomposition, precipitation, exothermic neutralisation, and displacement).

Question 2: Raju made the following observations during the reaction of some metals with dilute hydrochloric acid.

(a) Silver metal does not show any change.

(b) The temperature of the reaction mixture rises when aluminium (Al) is added.

(c) The sodium metal reaction is highly explosive.

(d) Some gas bubbles are seen when lead (Pb) is reacted with the acid.

Explain these observations giving suitable reasons.

Answer: (a) Silver metal does not show any change:

Reason: Silver (Ag) lies below hydrogen in the metal reactivity series and is less reactive than hydrogen. It cannot displace hydrogen from dilute hydrochloric acid.

(b) Temperature rises when Aluminium is added:

Reason: The reaction between aluminium metal and hydrochloric acid is highly exothermic ($2Al + 6HCl \rightarrow 2AlCl_3 + 3H_2$ + Heat). Energy released raises the temperature.

(c) Reaction with Sodium metal is highly explosive:

Reason: Sodium (Na) is an extremely reactive alkali metal at the top of the reactivity series. It reacts violently with acid, producing hydrogen gas and intense heat that ignites the gas explosively.

(d) Gas bubbles are seen when Lead reacts with acid:

Reason: Lead (Pb) lies above hydrogen in the reactivity series. It reacts slowly with dilute HCl to liberate hydrogen gas bubbles: $Pb(s) + 2HCl(aq) \rightarrow PbCl_2(s) + H_2(g)\uparrow$.

Explanation: Reactivity of metals with acids depends directly on their position relative to hydrogen in the reactivity series.